TEAM CHAKSHU

Unmanned Ground Vehicle



Team Members

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Our Capstone Mentors:-

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- **Dr. Dheeraj Gupta**
- **Dr. Ashish Singla**

EVEN SEM 2026 (JAN 2026 - JUN 2026)

Problem Statement

Problem	Limitations
• In forested, damp and high-slope terrains such as those found in North-East India, conventional mine detection methods show reduced reliability. • It is primarily due to soil variability, resulting in false alarms, delays, and increased risk to personnel.	• Existing solutions are either human-dependent, cost-prohibitive, or environment-specific • Limited availability of low-cost, modular UGVs suitable for Indian conditions

Objective:

To design and develop a UGV prototype for safe ground-based operations with real-time monitoring, obstacle avoidance, and terrain adaptability.

Target Audience/Collaborator



2 Kharga Corps



13 J&K Light Infantry



3 Kharga Corps



4 Kharga Corps



33 Kharga Corps



Assam Rifles



Border Security Force



Bordering State Police
(Eg. Punjab Police)

Questionnaire

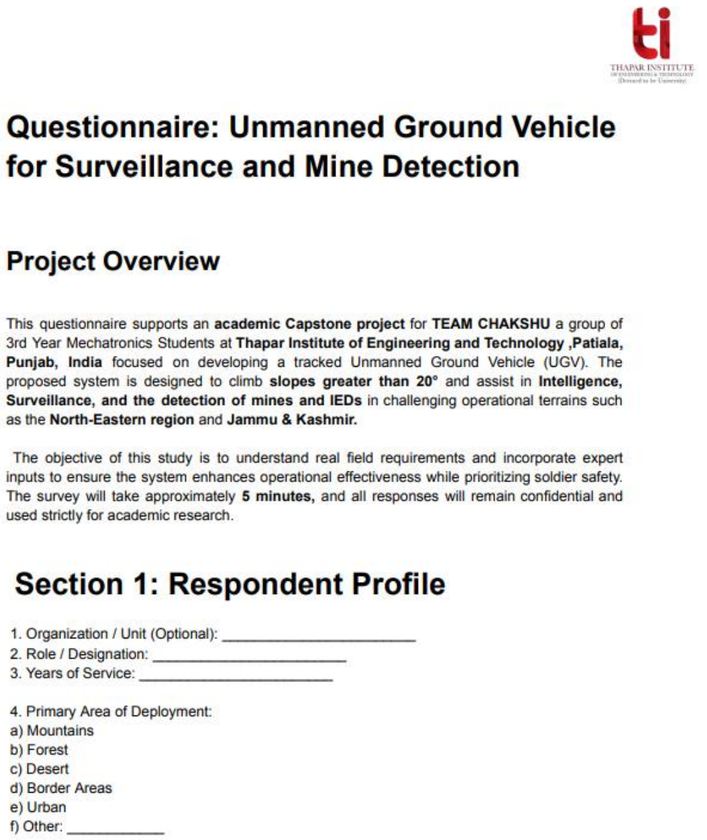
Questionnaire sent to different Armed forces contacts and Alumni contacts for better manufacturing and market understanding.

Alumni contacts :

1. **Mr. Ankit Sharma, R&D Head, SANDHAR Group**
2. **Mr. Subheg Singh, Business Development Manager, Gabriel India**

Armed Forces Contacts:

1. **Colonel, 2 Kharga Corps , Patiala**
2. **Lt. Colonel, 2 Kharga Corps, Patiala**
3. **Captain , 2 Kharga Corps, Patiala**
4. **Colonel, 13 J&K Light Infantry, Ambala**
5. **Army Design Bureau, Delhi**



THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

Questionnaire: Unmanned Ground Vehicle for Surveillance and Mine Detection

Project Overview

This questionnaire supports an **academic Capstone project** for **TEAM CHAKSHU** a group of 3rd Year Mechatronics Students at **Thapar Institute of Engineering and Technology ,Patiala, Punjab, India** focused on developing a tracked Unmanned Ground Vehicle (UGV). The proposed system is designed to climb **slopes greater than 20°** and assist in **Intelligence, Surveillance, and the detection of mines and IEDs** in challenging operational terrains such as the **North-Eastern region** and **Jammu & Kashmir.**

The objective of this study is to understand real field requirements and incorporate expert inputs to ensure the system enhances operational effectiveness while prioritizing soldier safety. The survey will take approximately **5 minutes,** and all responses will remain confidential and used strictly for academic research.

Section 1: Respondent Profile

1. Organization / Unit (Optional): _____
2. Role / Designation: _____
3. Years of Service: _____

4. Primary Area of Deployment:
a) Mountains
b) Forest
c) Desert
d) Border Areas
e) Urban
f) Other: _____

Questionnaire

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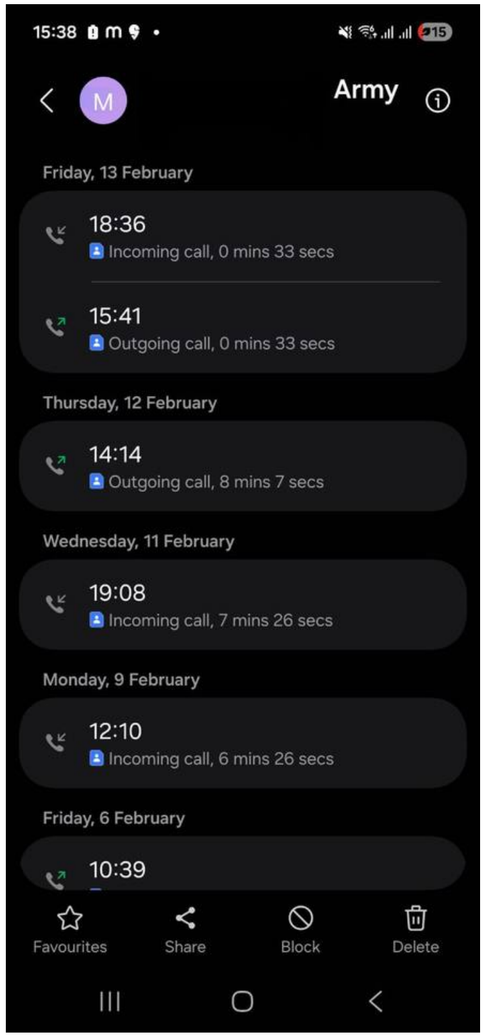
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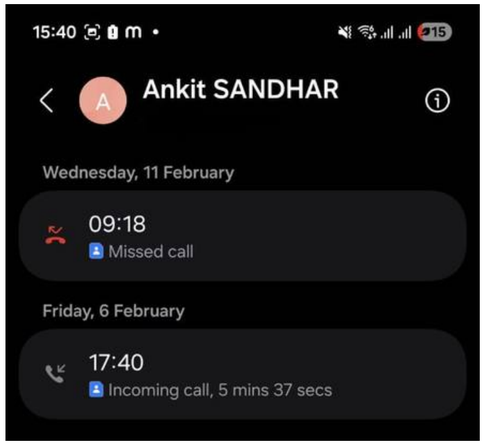
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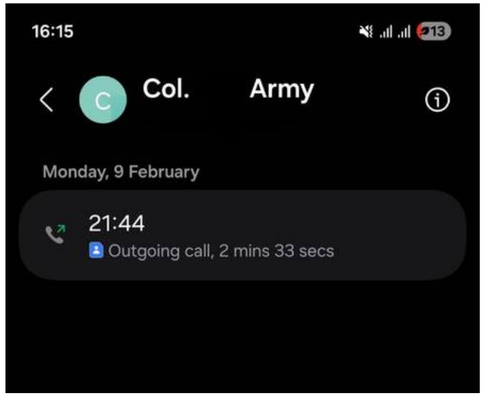
1. Organization / Unit (Optional): 2 Khaya Corps, 13 JRK Light Infantry
2. Role / Designation: Colonel, Captain, LoC
3. Years of Service: 24 yrs, 2 yrs, 26 yrs

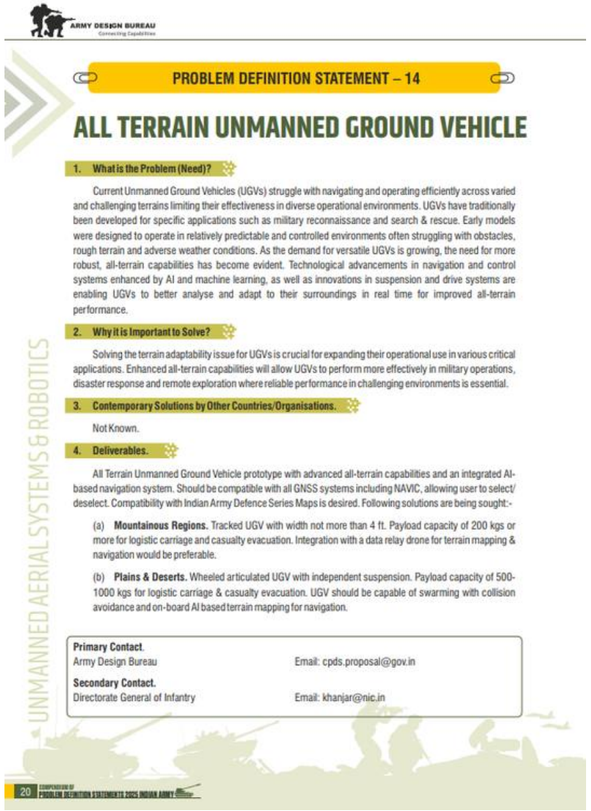
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f) Other: _____

[CAPSTONE RESPONSE .pdf]








ARMY DESIGN BUREAU

PROBLEM DEFINITION STATEMENT – 14

ALL TERRAIN UNMANNED GROUND VEHICLE

UNMANNED AERIAL SYSTEMS & ROBOTICS

1. What is the Problem (Need)?

Current Unmanned Ground Vehicles (UGVs) struggle with navigating and operating efficiently across varied and challenging terrains limiting their effectiveness in diverse operational environments. UGVs have traditionally been developed for specific applications such as military reconnaissance and search & rescue. Early models were designed to operate in relatively predictable and controlled environments often struggling with obstacles, rough terrain and adverse weather conditions. As the demand for versatile UGVs is growing, the need for more robust, all-terrain capabilities has become evident. Technological advancements in navigation and control systems enhanced by AI and machine learning, as well as innovations in suspension and drive systems are enabling UGVs to better analyse and adapt to their surroundings in real time for improved all-terrain performance.

2. Why it is Important to Solve?

Solving the terrain adaptability issue for UGVs is crucial for expanding their operational use in various critical applications. Enhanced all-terrain capabilities will allow UGVs to perform more effectively in military operations, disaster response and remote exploration where reliable performance in challenging environments is essential.

3. Contemporary Solutions by Other Countries/Organisations.

Not Known.

4. Deliverables.

All Terrain Unmanned Ground Vehicle prototype with advanced all-terrain capabilities and an integrated AI-based navigation system. Should be compatible with all GNSS systems including NAVIC, allowing user to select/ deselect. Compatibility with Indian Army Defence Series Maps is desired. Following solutions are being sought:-

(a) **Mountainous Regions.** Tracked UGV with width not more than 4 ft. Payload capacity of 200 kgs or more for logistic carriage and casualty evacuation. Integration with a data relay drone for terrain mapping & navigation would be preferable.

(b) **Plains & Deserts.** Wheeled articulated UGV with independent suspension. Payload capacity of 500-1000 kgs for logistic carriage & casualty evacuation. UGV should be capable of swarming with collision avoidance and on-board AI based terrain mapping for navigation.

Primary Contact.
Army Design Bureau — Email: cpds.proposal@gov.in
Secondary Contact.
Directorate General of Infantry — Email: khanjar@nic.in

Need Analysis

Army Wishlist

Terrain Challenge	Slopes, Riverbeds and Mud/Slush
Risks	Soldier loss, Detection delays
Minimal useful slope	20-25 degree
Preferred Maximum system weight	<10kg (Ideal), 10-20kg (Acceptable)
Critical Sensors	Day Camera, Live Video Transmission, GPR, Mine Detection Probe
Operational Range	500m -1km
Mission Endurance	2-4 hours
System Deployment Time	<2 minutes

Existing Solutions



CIRAKSHAKBOT



XPLODER



DOGO



VARAHA



CILYNX



L&T WHEELED UGV

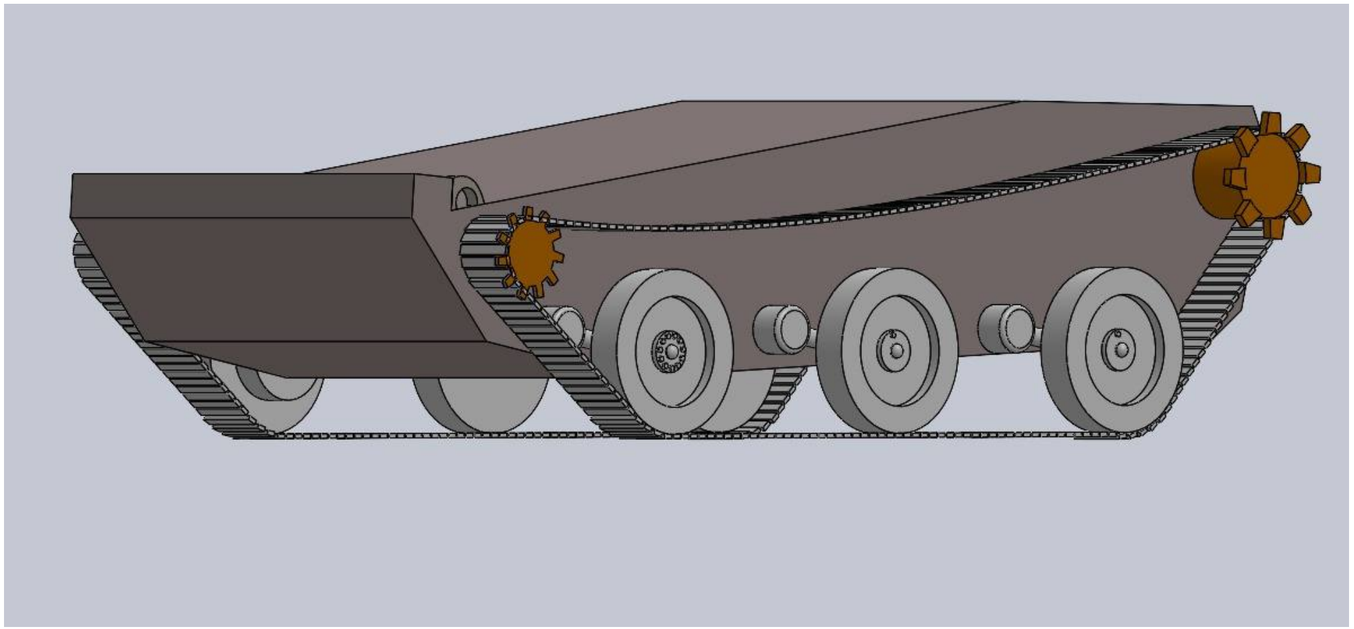
Existing/Our Solutions

Platform	Weight (kg)	Dimensions (mm approx)	Max Incline (°)	Endurance (hrs)	Max Payload (kg)	Mine Detection Capability	Technology Gap	Mobility Type	Operational Range
CIRAKSHAKBOT	15	805 × 470 × 305	~30+ (Stair capable)	2	30	Not integrated natively	No sensor fusion	Articulated Tracks	~800 m
CILYNX	4	386 × 356 × 216	~20–25	1–2	1.2	None	Tactical-only platform	Wheel/Track Hybrid	500–1000 m
VARAHA ROV	6	533 × 266 × 296	60	3	<3 (practical)	Optional Metal Detector	No area mine mapping	Wheeled	Short tactical
OUR UGV	≤20	1000 × 1000 × 500	≥30 (Design target 35°)	3–4	5–8 + Drone Dock	GPR + Metal Detector + Probe + AI Fusion	Not area-coverage optimized	Lightweight Tracked + Flippers	1000+ m

Limitations

Platform	Primary Limitation	Secondary Limitation	Our Counter
CIRAKSHAKBOT	No integrated mine detection stack	Only 2hr Endurance	Multi-layer detection + 4 hr Endurance
CILYNX	Very low payload	No Mine Detection	Full Detection Suite
XPLODER	Heavy platform	Not man-portable	≤ 20 kg portable tracked system with ≥25° slope capability

Preliminary Idea



Materials used :

Polycarbonate Sheet - Shell

Aluminium 5083 Square Profile - Inner Frame

(High-Density Polyethylene)HDPE Layer - Blast Protection

Dimensions:

LENGTH=750 mm

BREADTH =400mm

HEIGHT=280mm

Weight Expectation

<15kg

Climb Capabilities

>25 Degree slope (Upto 20% Moisture soil)

It will have probes for mine detection, which will penetrate the soil with a certain force and retract itself.

Material Selection & Cost Analysis

Chassis (Inner Frame)	Aluminum 5083 / 5086 – 50×50 mm T-square profiles	Lightweight yet rigid; prevents frame twisting during climbs	₹250 – ₹350 per meter
Drive Wheels & Idlers	EN8 (Medium Carbon Steel) – Solid machined discs / reinforced hubs	Higher strength and fatigue resistance, suitable for rough terrain	₹800 – ₹1,200 (Total)
Track System	Reinforced Rubber - 100 mm wide chain	Provides grip on damp (~20% moisture) soil	₹5,500 – ₹7,500
Blast Shield	HDPE (5 mm thickness)	Sacrificial baseplate protecting low-mounted motors from blasts	₹150 – ₹200 per kg
External Shell	Polycarbonate (3 mm thickness)	High impact resistance; protects electronics	₹120 – ₹150 per sq. ft
Fasteners & Couplings	Stainless Steel	Corrosion resistance; adds rotating mass	₹500 – ₹800

Total : ₹8,900 - ₹10,500

Project Execution Roadmap

Stage 1 – UGV Platform Development

- Design and fabricate a chassis
- Integrate motors, motor drivers, and the battery system
- Implement ESP32-based wireless control
- Test stability, payload capacity, and runtime

Outcome: Fully functional remote-controlled UGV

Stage 2 – Primary Detection Integration

- Integrate the metal detector module
- Add a magnetometer for magnetic anomaly detection
- Implement a real-time alert system
- Perform controlled sand-bed testing

Outcome: Reliable layered metal detection system

Stage 3 – Intelligence Enhancement (Optional Upgrade)

- Integrate a camera for visual monitoring
- Improve detection validation
- Optimize sensor filtering and noise reduction

Outcome: Reduced false positives & improved accuracy

Stage 4 – Controlled Mechanical Validation (Advanced)

- Integrate linear actuator & load cell
- Apply controlled force measurement
- Log and analyze ground resistance data

Outcome: Data-driven validation capability

Electronics Cost Estimation

STAGE 1

Component	Specification / Purpose	Estimated Cost (INR)
ESP32 Controller	Built-in WiFi & Bluetooth; Handles control and communication	₹400 – ₹600
Motor Driver (BTS7960 or Equivalent)	High current driver; Drives high torque motors; Handles current spikes	₹600 – ₹1,000
DC Geared Motors (4 Units)	High torque for rough terrain; Locally sourced for cost efficiency	₹4,000 – ₹6,000 (Total)
Li-ion Battery Pack (12V/24V)	Sized based on runtime and load requirement	₹2,500 – ₹4,000
Battery Management System(BMS) (Above Max Current Rating)	Battery protection and cell balancing	₹500 – ₹1,000
Buck Converter (5V Regulator)	Provides stable 5V supply for ESP32 and sensors	₹100 – ₹200

Total : ₹ 9,000-₹ 12,000

Electronics Cost Estimation

STAGE 2

Component	Specification / Purpose	Estimated Cost (INR)
Metal Detector Module (VLF / DIY Coil Based)	VLF-based sensing or custom coil design; Detects buried metallic objects; Primary detection layer	₹300 – ₹600
Magnetometer (HMC5883L or Equivalent)	3-axis magnetic field sensor; Detects magnetic field disturbance; Reduces false positives	₹150 – ₹300
Buzzer + LED Alert System	Audio-visual indicator; Provides immediate alert upon detection	₹50 – ₹100
Signal Conditioning & Filtering Components	RC filters / Op-amp based filtering; Improves detection accuracy; Reduces motor noise interference	₹100 – ₹200

Total : ₹1,200

References

THANK YOU